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A Method for Inter-organizational Business Process Management*

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Abstract -In this paper, an approach for inter-organizational business process management is introduced. The approach is backed by the UML modeling domain ontology among involved organizations and through the all phases in the business process management. XML Nets (a new variant of high-level Petri nets) is a formal, graphical modeling language that allows modeling both the flow of XML documents and the underlying business process. In the method, XML Nets combined with UML models for design, execution and monitoring of inter-organizational business processes. Consequently, the processes can be directly executed, improved and monitored. The approach supports the continuous improvement of inter-organizational business processes of all involved organizations on a holistic level.

Index Terms -Domain Ontology, Petri Nets, XML Nets, Business Process Management

I. INTRODUCTION

The emergence of e-Business emphasizes the importance of inter-organizational Business Process Management (iBPM). iBPM improves the competitive advantages of all involved organizations on a holistic level, but it requires the integration of the data of involved organizations, which should be exchanged automatically if required.

Domain ontology is used in the approach to enable and support the data integration from various systems regarding syntax, semantics and solving homonyms and synonyms problems.

XML Nets, a new variant of high-level Petri Nets, are a formal, graphical modeling language, which can be directly executed by a respective workflow engine and allows for analysis and simulation of distributed business processes. In order to improve the applicability of the approach, UML has been integrated. In this paper, XML Nets combined with UML models are applied to support all activities in BPM: process design, process execution and process monitoring (including process mining).

The paper is structured as follows: first we give a brief introduction to ontologies and UML, then modeling of a domain ontology with UML is described. In section 3, we introduce XML Nets. The transformation from UML models to Petri Nets Models is introduced in section 4, which is a preparatory work for the analysis of activities in BPM in section 5. Finally, we draw a conclusion of the work presented and give an outlook on future work.

II. Modelling A Domain Ontology with UML

A. Domain Ontology of BPM

In both computer science and information science, an ontology is a data model that represents a set of concepts within a domain and the relationships between those concepts. In the involved organizations, people have a different understanding about the terms used and their relations. The aim of a Domain Ontology is to provide a common understanding in the field by capturing the related domain knowledge and formally describing it. In 2004, the World Wide Web Consortium (W3C) finished its standardization work on the Web Ontology Language (OWL), thus laying the foundations for a wide-spread use of ontologies in business.. With OWL, relevant concepts of the domain (its terminology), their properties and instances can be defined.

B. UML

The Unified Modeling Language (UML) is OMG's most-used specification, and the way the world models not only application structures, behavior, and architectures, but also business processes and data structures. [9]

UML, along with the Meta Object Facility (MOF) also provides a key foundation for OMG's Model-Driven Architecture®, which unifies each step of development and integration in business modeling, through architectural and application modeling, for development, deployment, maintenance, and evolution.

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UML2 is independent of the methodology which is used for analysis and design. Regardless of the methodology that one uses, one can use UML2 to express the model.

C. Modelling a Domain Ontology with UML

As expatiated in [5], currently, visual modelling of ontologies is predominantly done with form- or tree-based tools, which are not appropriate to show the relationships. Since UML has its advantage of visualization, universality and reusing existing tools, tools provide an automatic translation from the visual model to the OWL syntax, so it is easier than manually writing an OWL-ontology. Among all UML diagrams, the class diagram is suitable to describe the static parts of ontologies. A class diagram gives an overview of a domain by showing its concepts and their attributes as well as the relationships among them. Furthermore, some common basic steps or the process unit that most often referenced by involved organizations in BPM (such as exception transaction workflows) also can be taken into account. This part can be modelled by Behaviour Diagrams.

Using UML to define the domain ontology has two parts of work: an Ontology Definition Meta-model and a UML Ontology Profile, as shown in Fig. 1. The Ontology Definition Meta-model defines a meta-model for an ontology. This meta-model is built on the Meta Object Faculty (OMG 2002), a model driven integration framework for defining, manipulating as well as integrating metadata and data in a platform independent way. The UML profile determines a visual UML syntax to model ontologies, which is a maximal reuse of UML2 features and OWL features with a large array of industrial tools available for UML and other related OMG standards for the purpose of ontology development. [6][5] The result of UML modelling can be translated into the OWL syntax, which is essentially an XML or XMI document. In this way, it can be manipulated by XML Nets.

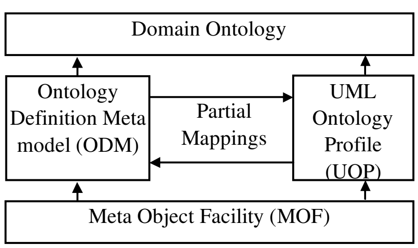

Fig. 1 Modelling the Domain Ontology with UML

III. XML Nets

XML Nets is a variant of high-level Petri nets, which is a formal, graphical modelling language that allows modelling both the flow of XML documents and the underlying business process. XML Nets is on the base the two techniques, namely GXSL (graphical XML schema definition language) and XManiLa (XML manipulating language). The detail definition and examples can be found in [2], the following just quote some to explain XML Nets.

A. GXSL(graphical XML schema definition language)

GXSL is a graphical XML schema definition language for the design of XML document types. It consists of a set of mark-up declarations of different types: element, attribute list, entity or notation declaration. GXSL is DTD-based, i.e. a DTD can be unambiguously derived from the graphical XML schema definition language (GXSL). [2]

Fig.2 shows a simple example of a Supplier XSD constructed in GXSL. Here omitted the textual representation, because it is easy to image. The graphical representation gives a better understandable and more direct overview of the defined elements and the logical structure of a valid supplier XML scheme.

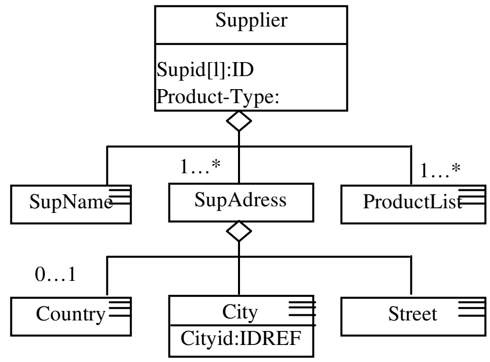

Fig. 2 XSD for the Supplier DTD

B. XManiLa (XML manipulating language)

XManiLa is the GXSL based XML document manipulation language, which can use GXSL with some extensions also for querying and manipulating XML documents. A GXSL-schema can be seen as a template for a set of XML documents that specifies the structure of matching documents. In order to enable also a content based query, we allow for assigning constants or variables to an element or an attribute. [2]

For example, a query can be composed as all the supplier that Product-Type= N and the value of the variable "Product" in "ProductList". The XSD showed in Fig. 3 represents this query.

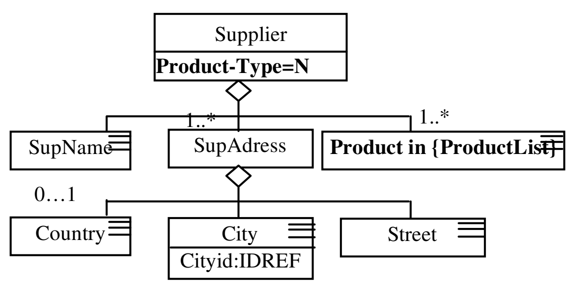

Fig. 3 Supplier that has the special product type and the specific product

C. XML Nets

In the above sections, how to graphically model XML DTDs with GXSL and how to specify document retrieval and document manipulation with XManiLa have been described. Combine both techniques for the definition of XML nets

The static components of ***XML nets*** (i.e. the places of the Petri net) are typed by XML schema diagrams, each of them representing a DTD. Places can be interpreted as containers for XML documents which are valid for the corresponding DTD.

The flows of XML documents are defined by the occurrences of transitions, which thereby manipulates (creates, changes or deletes) the documents of their adjacent places. The activation of a transition, which is prerequisite for the transition's occurrence, depends on the labels of the adjacent edges constructed with ***XManiLa*** and on an optional transition inscription.

We illustrate a fragment of a simple example business process; it is "when a specific product stock was checked, if it is below the certain "SecureValue", the purchase was needed. So before purchasing, need to choose the qualified supplier". Fig.4 shows this process fragment. In this process, mainly two business objects are involved, i.e. Supplier and Product,

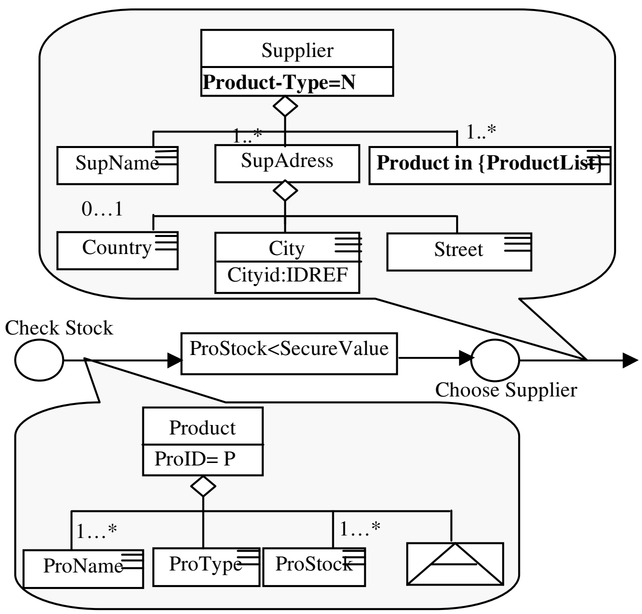

Fig. 4: Business Process Management

In XML Nets, the graphical description of XML Scheme and the manipulation of documents along with their underlying business objects are integrated with the Petri Net formalism.

When using XML Nets to model inter-organizational processes, the predefined relevant business objects can be modeled as the places, the static components of XML nets. These places can be considered as repositories for sets of predefined business object, whose structure can be represented by an XML-Schema. These schemas can identify all objects that a specific place can contain.

Transformations of one or more of these objects, performed by inter-organizational processes, are represented by the transitions. In the process schema, the transitions manipulate (create, change or delete) XML documents, which represent the underlying corresponding business objects. A process is therefore modelled by a manipulation of an XML document, which representing the real business objects.

Associated places and transitions are connected by arcs, which display the direction of the object flow, to which the manipulation filter is assigned.

It should be emphasized that – in contrast to other high-level Petri-nets – not only variables but also the hierarchical structures of objects can be manipulated. The filter can be considered as a template, which defines the set of tokens that can be manipulated by a transition. A transition is enabled if all places in its pre-set contain at least one token, which complies with the filter, assigned to the adjacent arc. Furthermore, optional specific inscriptions can be assigned to the transitions. These so-called transition inscriptions represent a set of logical expressions for certain variables of the documents, stored in the places of the pre-set. They can also include functions, which determine the values of variables and the structure of the tokens in the post-set after firing the transition. If an inscription is assigned to a transition, the transition is enabled only if all places in its pre-set contain at least one token which complies with the filter and all logical expressions have the value "true".

IV. Convert UML Models to Petri Nets Models.

A. Motivation

UML is a well known domain object-oriented modeling language for the design process of IT systems. However, despite of its success as unified and visual notation widely used in industry, UML still lacks of dynamic analysis, verification, and precise formal semantics capabilities, which hinder the formal validation of system design. On the other hand, models established in many mathematical model languages (such as Petri Nets) are precise and can be analyzed and verified by using various tools. Thus, in our approach, UML models that need to be analyzed and verified should be transformed into Petri Net models. Consequently, it can be easily integrated in XML Nets that support all activities in BPM. Much research work combined the UML and Petri Nets in order to gain the advantage of understandability, object-orient characteristic, universality of UML and the formalization, precise, behavioral modeling, analysis, the executable capability of Petri Nets.

B. Related work

The way of combination of UML and Petri Nets can be categorized into two. One is combing the UML and its object-oriented idea with Petri Nets, without transforming UML Diagrams to Petri Nets, to describe all relevant aspects of business processes, such as Object Coordination Nets (OCoNs) [10].

Another way has more high integration degree by transforming UML models into Petri Net models. A lot of successful research has been done to transfer the main kinds of UML diagrams into Petri Nets (including use case diagrams, object diagrams, class diagrams, activity diagrams, state diagrams, sequence diagrams and deployment diagrams) and many related tools have been developed to automatically support this transformation or to support the editing of both UML and Petri Nets. [3] [11] [12] [13] [14] [16] [22] [18] [21]

The second way differs from the aspects of the class of Petri Nets, the way they transform and the type of verification and validation. Among them, [12] has advantages of the clear structure, the object-oriented and extensible ability, which can fulfil our requirement. The approach in [12]([12] worked on the basis of [18]) based on a technique which derives colored Petri Nets from UML objects, state chart collaboration and sequence diagrams, which associates the formalization of dynamics and the behavior of objects identified by identities and attribute values. The details and a case study can be found in [12] [18]. Here we only use that method to support our method. After the UML model was transformed, it can be seamlessly integrated into XML Nets.

VI. Activities in Business Process Management supported by XML Nets

The activities which constitute business process management can be grouped into three categories: design, execution and monitoring. In the process design phase, there are two or three steps, if the design starts in UML modeling, there are three steps: UML modeling, XML Nets Modeling and model analysis. If it starts in XML Nets Modeling, there are two steps. And Domain Ontology supports all steps of the process design as shown in Fig. 5.

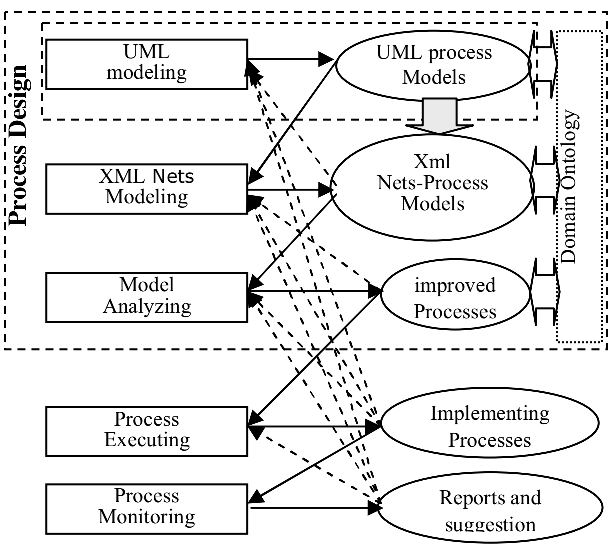

Fig. 5: Business Process Management

A. Process Design

UML is a standard modeling language, so it could be a good means when acquiring the business processes from both the professionals and non-professionals. In fact, there are many designers who prefer to model with UML. As described in section 4, the UML Models can be transformed into Petri Net models through the method bring forward in [12][18] . So it can be integrated in the following steps of BPM. In addition, a lot of systems left behind and systems from the other involved organizations were modeled by UML, so it is not advisable to ignore the UML modeling.

Some professionals certainly can start with XML Nets model to model a business process or part of the business process, such like the research in [2] which can be directly simulated and validated.

Because of the machine readable ability of XML Nets, a so-called simulation engine can interprets the formal syntax and semantics of XML Nets. Simulation of XML Nets allows for business process analysis on a very detailed level. The information got from the simulation can be used to improve the business processes. [1]

B. Process Execution

XML Nets can be directly executed by respective Workflow engines [2]. In the process execution, the process designed (or transferred form UML) by XML Nets are implemented automatically, this way of directly executing a process is much more straightforward and therefore easier to be analyzed and improved.

The system will use services in connected applications to perform business operations (e.g. calculating a repayment plan for a loan), which make the structure of the system more simple. Anyway the graphical process model of XML Nets is better than a text- based process model to understand, to analyze, to communicate and to diagnose.

C. Process Monitoring

In order to monitor the states of the business process, the performance indicators are defined and are represented by places in XML Nets. These performance indicators are mainly used to ascertain the economic efficiency of the business (such as time value, cost etc.) and the degree of the harmonization among the involved organizations in the supply chain. Performance indicators map the business circumstances onto quantitative values (e.g., system status etc.). Normally the certain target values or target values scope are predefined.

The process monitoring encompasses the tracking of individual processes so that information on their state and the provision of statistics can be easily received and collected. So the performance indicators can determine the management objectives and verify the implementation of those objectives (i.e., monitoring variance between current and target values) to check the states of the system.

The calculation of a performance indicator value can be modeled by parts of a XML Net and supports a direct monitoring of the inter-organizational processes in all phases of business process management and not only in the ex post view. In addition, this information can be used to guide customers and suppliers to improve their connected processes and their cooperation.

Process mining can be done by assigning some predefined values or ranges to the performance indicators, whose aim is to analyze event logs extracted through process monitoring and to compare them with an 'a priori' process model aided by some tools. Process mining allows process analysts to detect discrepancies between the actual process execution and the a priori model as well as to analyze bottlenecks, and then to solve it.

VII. Conclusions

In this contribution we have introduced an approach for the integration of UML models and XML Nets models to support all activities in the inter-organizational Business Process Management.

In the design phase, XML Nets model was built and also UML can be built, which afterward was transferred to the Petri Nets model , so can be integrated with the XML Nets. The graphical result of this design phase is standard and easy to understand, which reduce the complexity of the inter-organization business process.

These XML Nets model is machine readable, so it can be automatically executed in the implementing phase by the respective Workflow engines, which is more directly.

Using Xml Nets, the performance Indicators can be set to monitoring the system performance, consequently check the states of the business process and to improve it.

We used an industry standard as an object-oriented modeling language, UML plus the behavioral modeling, analysis and simulation strengths of Petri Nets, and the UML modeling domain ontology was used to acquire a common understanding about the domain terminology for all involved organizations. Future research will focus on the implementation of the presented approach.

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